



Markov Chain Analysis of Gaze Transitions: Assessing Attention Management in SAE Level 3 Automated Driving

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ABSTRACT

This work-in-progress examines how the use of different non-driving related task (NDRT) interfaces (mobile phone vs. head-up display and baseline) influences drivers' visual attention and scanning strategies during conditional automated driving. We present preliminary findings from a driving simulator study (N=46) that utilised Markov Chain analysis of gaze transitions and gaze dispersion metrics. Results show that NDRTs, particularly on mobile phones, compromise drivers' attention distribution and gaze transitions. While HUDs keep drivers' gaze closer to the road, they may still hinder hazard perception. Markov Chain analysis reveals valuable insights into drivers' attention management, informing the design of safer in-vehicle interfaces for automated vehicles. These findings highlight the need for careful consideration when designing in-vehicle interfaces and NDRT interactions for automated vehicles.

CCS CONCEPTS

• **Human-centered computing** → **User studies; Laboratory experiments; HCI theory, concepts and models.**

KEYWORDS

NDRT; Gaze Behaviour; Automated Vehicles; HMIs; Markov Chain Analysis

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1 INTRODUCTION

As vehicle automation advances, drivers are more likely to engage in non-driving related tasks (NDRTs) [1], which may compromise their situation awareness [4, 15] and ability to resume control when needed [2, 13]. This issue is particularly critical for SAE levels 2/3 automated vehicles (AVs)[16], where drivers must be able to safely recover control if the vehicle encounters operational design domain (ODD) limitations. Effective attention management, such as adequately splitting attentional resources between the driving task and NDRTs, is crucial for maintaining situation awareness [18, 22] and affects take-over performance during safety-critical scenarios [9, 11, 12, 14, 20].

Previous studies on this topic [6, 10] have primarily relied on gaze concentration metrics to assess drivers' visual attention, providing insights into overall attention distribution patterns. However, these measures have limited capabilities in explaining safety-related scanning patterns and attention shifts, which are relevant in complex driving scenarios. Examining attention shifts can reveal how quickly and efficiently drivers re-focus their attention on critical

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road areas when necessary, offering additional insights into scanning behaviour and potential safety implications.

Markov Chain analysis has emerged as a promising tool for understanding drivers' gaze transitions and attention management strategies in various driving contexts. It has been successfully applied to investigate drivers' hazard perception scanning patterns in both manual driving and vehicle automation [see 7, 19]. This approach models the probability of a driver's gaze transitioning from one area of interest to another based on their current gaze location. By considering the temporal aspects of gaze transitions, Markov Chain analysis can provide valuable insights into drivers' attention management strategies and identify patterns linked to safety-relevant behaviours and potential crash incidents [19].

While previous studies have applied Markov Chain analysis to investigate drivers' hazard perception in manual driving and vehicle automation, its potential for examining attention management in the context of NDRTs and conditionally automated vehicles remains largely unexplored. In light of this research gap, we present the application of Markov Chain analysis in this context. Our approach seeks to provide novel insights into the utility of Markov Chain analysis, as well as the temporal aspects of drivers' attention management strategies, complementing traditional gaze concentration metrics.

2 METHODOLOGY

We used an existing driving simulator study to showcase the application and utility of Markov Chain analysis in understanding drivers' attention management strategies in conditionally automated vehicles with NDRTs—in particular, insights into the temporal aspects of drivers' attention allocation and the potential to inform the future design of AVs interfaces. Below, we first describe the simulator study for context, followed by a description of the Markov Chain analysis.

2.1 Simulator Study

The dataset underlying the Markov Chain analysis is representative of other conditionally automated driving simulator studies, including a driving simulator with eye-tracking hardware with a range of defined areas of interest (AoIs), NDRTs (in this case, watching videos on different display modes: mobile phone vs. head-up display and baseline) and a driving scenario with automated driving and a take-over request (TOR).

2.1.1 Experiment Apparatus. The study employed an advanced driving simulator with a real-vehicle cabin, a surround-sound system, and a six-degree-of-freedom motion platform. Three front-view projection screens and LCD monitors for side and rear-view mirrors provided the driver with a 180-degree high-resolution field of view (cf. Figure 1).

Eye-tracking data were collected using the Driver Monitoring System (PCDMS) developed by Seeing Machines Limited, which features an infrared camera and pulsed infrared lights. The PCDMS monitored drivers' head and eye movements at a sample rate of 46 Hz during the automated drive (approximately 13 minutes). The extracted measurements included gaze yaw, gaze pitch, and the intersection of drivers' gaze with 10 predefined AoIs.

Driving scenarios were created using real-world recorded videos of a manually driven car mimicking an automated vehicle's performance [5, 8]. Three similar 15-minute trips were selected, each including inner-city roads, a major motorway, and suburban roads, with a simulated planned TOR issued at the exit of the motorway.



Figure 1: Advanced Driving Simulator.

2.1.2 Participants. Forty-six participants (23 males, 23 females; mean age 32.3 years, SD = 8.7) with valid driver's licenses completed the study. Through social media posts, university website advertisements, and snowball sampling, participants were recruited from local communities.

2.1.3 Study Design and Procedure. The study employed a within-subject design in which all participants underwent three 13-minute drives in an SAE Level 3 AV [16] simulator. Each drive, which was counterbalanced across the scenarios, included a planned TOR when exiting the motorway on the same route, and one of three counterbalanced NDRT engagement conditions: watching a video on a mobile phone, watching a video on a head-up display, and a baseline condition with no video. Drivers were allowed to position the mobile phone at any position they felt comfortable. Figure 2 displays the mobile and HUD conditions. The video was only shown during the automated driving phase. Lastly, a planned TOR was issued towards the end of the driving route when the vehicle exited the motorway.

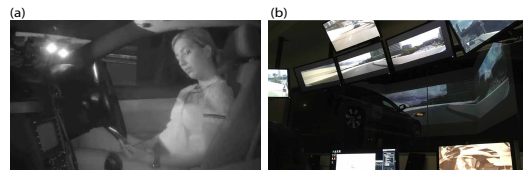


Figure 2: NDRT display modes: (a) Mobile condition, showing participant watching video on their handheld device, and (b) HUD condition, where the HUD to watch the video is projected with 61.75% transparency on the centre screen.

The entire experiment took approximately 1 hour, and participants received a \$70 gift voucher for their participation. The study was conducted in accordance with the national ethics code and approved by the university's ethics review board (approval number 1700000425).

2.2 Markov Chain Analysis

Our comprehensive analysis of driver gaze behaviour under three conditions—Baseline, Mobile, and HUD—utilised a multifaceted approach to examine how each interface affects attention management and gaze scanning strategies during the automated drive.

It is important to underscore that the analysis only covered drivers' gaze during the periods that the automation was engaged (before the TOR was issued). This study focused specifically on drivers' attention management strategies during the automated driving phase. The relationship between these strategies and take-over performance, though important, is beyond the scope of this work-in-progress. Prior to the analysis, the collected gaze data were filtered and validated to ensure that the "Unknown" AoI represented gaze fixations outside the predefined AoIs and did not include any missing data. The primary method employed in this analysis was Markov Chain analysis, which was used to model and understand drivers' gaze transitions between different AoIs.

The Markov Chain analysis generated a series of gaze fixation transition matrices for each participant, which contain the probability for each transition of fixations between AoIs to happen. These matrices were used to compare the fixation transition probabilities across conditions. By applying Markov Chain analysis to drivers' gaze transitions, we aimed to gain insights into their attention allocation and scanning patterns under different NDRT conditions. To complement our analysis, Markov Chain analysis (i.e. gaze transition matrices) were used in parallel with drivers' vertical/horizontal gaze dispersion.

Fixations were calculated as a point of permanence of drivers' gaze exceeding a minimum of 150 ms under a distance threshold of a 1-degree radius. The technique and thresholds for fixation detection are in line with standard protocols used in the field [17] and consistent with other studies in the field of gaze behaviour in automated driving [7, 12]. Gaze dispersion was calculated as the standard deviation of drivers' gaze positions throughout their automated driving periods.

Diverse statistical tests were used to compare the drivers' gaze behaviour differences between the experimental conditions. Shapiro-Wilk tests were used to assess the dataset's normality. For cases where normality was assumed, parametric statistical tests were conducted using ANOVA tests. Since the gaze transition probabilities generated by the Markov Chain analysis failed the normality tests, non-parametric Wilcoxon's tests were used for statistical evaluation. An alpha value of 0.05 (95%) was used for all tests to assess the statistical significance. Individual comparisons between scenario conditions were evaluated by Bonferroni post-hoc tests. For the cases where non-parametric tests were applied, sample-paired Mann-Whitney U tests were used as post-hoc tests.

3 RESULTS

3.1 Gaze Dispersion

ANOVAs revealed significant effects of the experimental conditions on drivers' vertical [$F(2, 137) = 127.21, p < 0.001, \eta_p^2 = 0.635$] and horizontal gaze dispersion [$F(2, 137) = 31.943, p < 0.001, \eta_p^2 = 0.321$]. The mobile condition showed the highest vertical dispersion ($M = 10.913^\circ, SD = 3.070^\circ$), while the HUD condition had the lowest horizontal dispersion ($M = 9.628^\circ, SD = 4.585^\circ$) (cf. Figure 3).

3.2 Gaze Fixations Transitions

Aggregated transition matrices (cf. Figure 4), computed by the Markov Chain analysis, and Wilcoxon tests (cf. Table 1) revealed

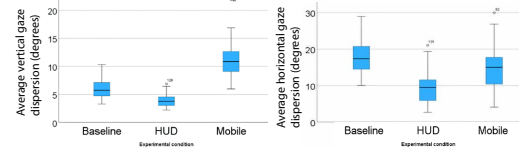


Figure 3: Average vertical (left) and horizontal (right) gaze dispersion for the three conditions.

that drivers in the mobile condition had a lower probability of transitioning their gaze towards the "Centre Screen" AoI and a higher probability of transitioning towards the "Unknown" AoI compared to the baseline and HUD conditions. HUD and baseline conditions had similar proportions of gaze transitions towards the "Centre Screen", but HUD drivers were more likely to transition from the "Centre Screen" to the "Unknown" AoI. Baseline drivers had a significantly higher proportion of gaze transitions from the "Centre Screen" to the "Left Screen" and "Right Screen".

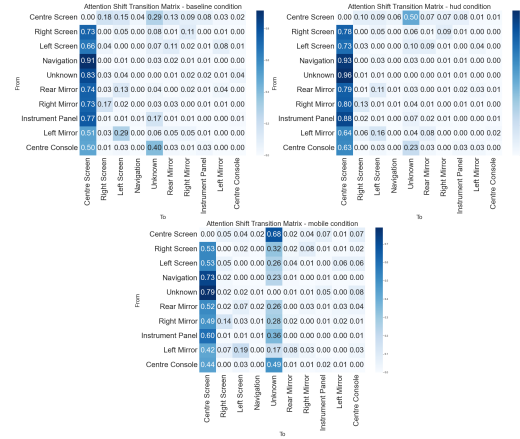


Figure 4: Aggregated transition matrices: average gaze transition probability for all participants per condition, baseline (top), HUD (middle) and mobile (bottom).

4 DISCUSSION

Our analysis of drivers' visual attention in an SAE level 3 automated vehicle revealed that both NDRT conditions (HUD and mobile) significantly impacted drivers' gaze patterns. Both reduced lateral scanning and functional field of view, with the mobile condition causing higher vertical gaze dispersion away from the road.

The Markov Chain analysis revealed that drivers in the mobile condition had a higher probability of shifting their gaze to the "Unknown" AoI and were less likely to return their gaze towards the "Centre Screen" after fixating on any AoI not related to the road environment. This suggests that handheld mobile NDRTs increase variability in drivers' attention shifts, potentially compromising the occurrence of relevant gaze transitions for hazard perception and reducing their take-over readiness due to reduced situation awareness [3, 12, 21].

Table 1: Wilcoxon Tests

ORIGIN	Navigation	Unknown	Centre Screen					Instruments Panel		Left Screen	Rear Mirror	Right Mirror	Right Screen
DESTINATION	Centre Screen	Centre Screen	Unknown	Left Screen	Rear Mirror	Right Mirror	Right Screen	Unknown	Centre Screen	Centre Screen	Centre Screen	Centre Screen	Centre Screen
N	132	132	132	132	132	132	132	132	132	132	132	132	132
Combined Median	0.725	0.857	0.4365	0.075	0.029	0.041	0.08	0.138	0.667	0.6325	0.581	0.571	0.698
Mean Baseline	0.69823	0.81932	0.27507	0.16082	0.11398	0.09407	0.18584	0.20198	0.68673	0.66211	0.656	0.64402	0.73132
Mean HUD	0.54525	0.88123	0.47239	0.0885	0.08527	0.07507	0.09655	0.09545	0.60343	0.66336	0.58759	0.5532	0.60384
Mean Mobile	0.37866	0.75227	0.66014	0.04914	0.04914	0.01402	0.03498	0.37827	0.46018	0.36375	0.32395	0.32118	0.42002
Chi-Square	16.545	24.43	40.909	32.909	26.727	10.972	30.977	26.727	12.13	20.364	13.636	19.701	16.545
df	2	2	2	2	2	2	2	2	2	2	2	2	2
Sig.	>0.001	>0.001	>0.001	>0.001	>0.001	0.004	>0.001	>0.001	0.002	>0.001	0.001	>0.001	>0.001

* note that only the significant results are being reported

*note that only transitions with the combined mean higher than 5% (0.05) of occurrence was reported

Video analysis revealed that most drivers positioned their phones in locations that diverted their gaze from the road: 71% near the steering wheel and 28% on their lap. These locations fall within the "Unknown" AoI, indicating that drivers may not be attending to important driving-related information such as the road ahead, mirrors, or instrument panel. Thus, the mobile device location would be classified inside the "Unknown" AoI, according to the PCDMS world model.

Although the HUD condition kept drivers' gaze closer to the forward roadway, the Markov Chain analysis showed that drivers in the HUD condition were less likely to perform eventual checks on safety-relevant areas of interest compared to the baseline condition when deviating their gaze away from the "Centre Screen". This suggests that the HUD may lead to a narrower attentional focus on the NDRT, limiting drivers' ability to effectively scan the driving environment.

Markov Chain analysis provided a nuanced understanding of drivers' attention management strategies that gaze dispersion measurements alone couldn't capture. offers unique and valuable insights into drivers' attention management strategies [see 7, 19]. These findings underscore the importance of considering detailed attention management when developing in-vehicle interfaces for conditionally automated vehicles, potentially informing the design of safer and more effective NDRTs.

5 CONCLUSION AND FUTURE WORK

Markov Chain analysis of gaze transitions offers a promising approach to understanding drivers' attention allocation and informing the design of safer in-vehicle interfaces. By identifying the proportions and probabilities of gaze transitions between different areas of interest, this approach provides a nuanced and interpretable understanding of drivers' gaze behaviour and attention management strategies.

It is important to acknowledge the limitations of the current study. The findings are based on a simulator experiment, and while

the advanced driving simulator used in this study provides a high level of realism, it may not fully capture the complexities of real-world driving scenarios. Additionally, the study focused on a specific type of NDRT (watching a video) and a single level of vehicle automation (SAE Level 3). Future research should investigate the generalizability of these findings to a wider range of NDRTs, such as reading, typing, or engaging in conversation, as well as different levels of vehicle automation, including SAE Levels 2 and 4.

Furthermore, future research should investigate optimal NDRT placements and the integration of driver monitoring systems to support safe interactions with NDRTs in AVs. The identified gaze transition patterns computed by Markov Chain analysis could guide the placement of critical information on HUDs and prompt drivers to scan safety-relevant areas periodically. Additionally, the potential generalisability of these findings to real-world driving scenarios and different levels of vehicle automation should be explored. Lastly, the potential long-term effects of NDRT engagement on drivers' attention management strategies and the development of adaptive in-vehicle interfaces that respond to changes in driver attention should be explored.

The preliminary findings of this study have significant implications for the AutomotiveUI community. Our research underscores the importance of carefully designing in-vehicle interfaces and NDRT interactions in conditionally automated vehicles. By demonstrating the impact of NDRT display modes on drivers' attention management, we highlight the need to minimise potential negative effects on attention allocation and hazard perception capabilities. These preliminary findings contribute to the growing body of research on drivers' attention management in conditionally automated vehicles.

Our study demonstrates the value of Markov Chain analysis in understanding drivers' gaze transitions and attention management strategies, opening up new avenues for research and development in the field of automotive user interfaces. The findings from our approach go beyond the acknowledgement of drivers' attention focus to the NDRT through observation of their gaze dispersion. The

insights from this study may inform how new technologies change drivers' scanning strategies and provide a more comprehensive link to their hazard perception capabilities. Furthermore, these findings may contribute to the design of adaptive, user-centred and context-aware interfaces that promote safer and more effective interactions between drivers and automated vehicles. Ultimately, our research contributes to the broader goal of the AutomotiveUI community in creating user-centred, safe, and efficient interfaces for the future of automated driving.

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